

Microsoft Intune + Graph Automation Lab (Classroom Guide)

Goal

Learn how to:

1. Install PowerShell 7
 2. Install Microsoft Graph PowerShell
 3. Connect to Microsoft Graph
 4. Create user via automation
 5. Assign Intune license
 6. Verify licensing
-

PART 1 — Install PowerShell 7

Download

<https://aka.ms/powershell-release>

Open PowerShell 7

Open Start Menu:

pwsh

PART 2 — Install Microsoft Graph Module

Install module

```
Install-Module Microsoft.Graph -Scope CurrentUser -Force
```



PART 3 — Connect to Microsoft Graph



Stable Connection Method

```
Connect-MgGraph `
  -Scopes "User.ReadWrite.All","Directory.ReadWrite.All" `
  -UseDeviceAuthentication `
  -ContextScope Process
```



Authentication

1. Browser opens OR device code appears
2. Open:

`https://login.microsoft.com/device`

3. Enter device code
4. Login using admin account:

`admin@<tenant>.onmicrosoft.com`



PART 4 — Verify Connection



Check session

```
Get-MgContext
```



Test Graph access

```
Get-MgUser -Top 3
```



PART 5 — Create User via Automation



Create password profile

```
$PasswordProfile = @{
  Password = "TempP@ss1234"
  ForceChangePasswordNextSignIn = $true
}
```



Create user

```
New-MgUser `
  -DisplayName "Automation User 1" `
  -UserPrincipalName "autouser1@<tenant>.onmicrosoft.com" `
  -MailNickname "autouser1" `
  -AccountEnabled `
  -PasswordProfile $PasswordProfile
```



PART 6 — Verify User Creation

```
Get-MgUser -Filter "startswith(DisplayName,'Automation')"
```



PART 7 — Set Usage Location (MANDATORY)



Required before license assignment

```
Update-MgUser `
  -UserId "autouser1@<tenant>.onmicrosoft.com" `
  -UsageLocation "IN"
```



PART 8 — Get Available Licenses

```
Get-MgSubscribedSku | Select SkuId, SkuPartNumber
```



Find:

```
INTUNE_A
```

This is:



Microsoft Intune Plan 1



PART 9 — Save Intune SKU

Example:

```
$skuId = "061f9ace-7d42-4136-88ac-31dc755f143f"
```



PART 10 — Assign Intune License

```
Set-MgUserLicense `
  -UserId "autouser1@<tenant>.onmicrosoft.com" `
  -AddLicenses @{SkuId = $skuId} `
  -RemoveLicenses @()
```



PART 11 — Verify License Assignment

```
Get-MgUserLicenseDetail `
  -UserId "autouser1@<tenant>.onmicrosoft.com"
```



Expected Result

```
SkuPartNumber : INTUNE_A
```



PART 12 — Remove Wrong License (Optional)

Example: Remove Intune Suite

```
Set-MgUserLicense `
  -UserId "autouser1@<tenant>.onmicrosoft.com" `
  -AddLicenses @() `
  -RemoveLicenses @("a929cd4d-8672-47c9-8664-159c1f322ba8")
```



PART 13 — Reconnect Session (If Needed)



If Graph session breaks

Example error:

```
DeviceCodeCredential authentication failed
```



Reconnect cleanly

Disconnect-MgGraph

```
Connect-MgGraph `
  -Scopes "User.ReadWrite.All","Directory.ReadWrite.All" `
  -UseDeviceAuthentication `
  -ContextScope Process
```



ARCHITECTURE UNDERSTANDING

◆ Services Used

Service	Purpose
Microsoft Entra ID Identity	
Microsoft Intune	Device management
Microsoft Graph	Automation/API



Flow

```
PowerShell
  ↓
Microsoft Graph
  ↓
Entra ID / Intune
```



IMPORTANT LEARNING

SCCM Automation

Get-CMDevice

Uses:

👉 Configuration Manager provider

Intune Automation

Get-MgUser

Uses:

👉 Microsoft Graph API

BEST PRACTICES

Always

- Use PowerShell 7
 - Use `-ContextScope Process`
 - Verify every step
-

Avoid

- PowerShell 5.1
 - Skipping UsageLocation
 - Assigning wrong SKU
-

FINAL RESULT

You now know how to:

- ✓ Connect to Graph
 - ✓ Create users
 - ✓ Assign Intune licenses
 - ✓ Verify automation
 - ✓ Understand Graph architecture
-

NEXT PHASE

After this:

- Device enrollment
- Intune policies
- App deployment
- Compliance policies

- SCCM + Intune co-management
-